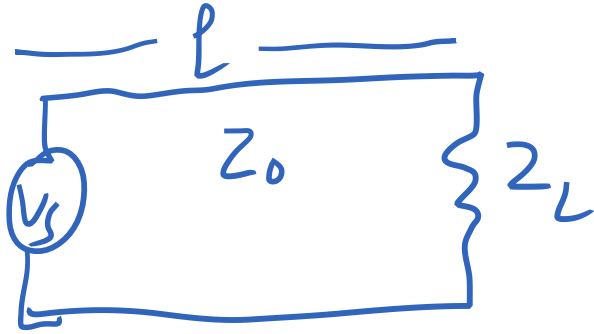


Wave characteristic of finite Transmission line



$$Z(l) = Z_0 \left[\frac{1 + \Gamma(l)}{1 - \Gamma(l)} \right]$$

$$\Gamma(l) = \Gamma_0 e^{-2\gamma l}$$

$$\Gamma_0 = \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right)$$

$$Z(l) = Z_0 \left[\frac{1 + \frac{Z_L - Z_0}{Z_L + Z_0} e^{-2\gamma l}}{1 - \frac{Z_L - Z_0}{Z_L + Z_0} e^{-2\gamma l}} \right]$$

Wave characteristic of finite Transmission line



$$Z(l) = Z_0 \left[1 + \frac{Z_L - Z_0}{Z_L + Z_0} e^{-2\gamma l} \right] \left[1 - \frac{Z_L - Z_0}{Z_L + Z_0} e^{-2\gamma l} \right]$$

$$Z(l) = Z_0 \left[\frac{(Z_L + Z_0) + (Z_L - Z_0) e^{-2\gamma l}}{(Z_L + Z_0) - (Z_L - Z_0) e^{-2\gamma l}} \right]$$

$$Z(l) = Z_0 \left[\frac{Z_L (e^{\gamma l} + e^{-\gamma l}) + Z_0 (e^{\gamma l} - e^{-\gamma l})}{Z_L (e^{\gamma l} - e^{-\gamma l}) + Z_0 (e^{\gamma l} + e^{-\gamma l})} \right]$$

Wave characteristic of finite Transmission line



$$Z(l) = Z_0 \left[\frac{Z_L (e^{\gamma l} + e^{-\gamma l}) + Z_0 (e^{\gamma l} - e^{-\gamma l})}{Z_L (e^{\gamma l} - e^{-\gamma l}) + Z_0 (e^{\gamma l} + e^{-\gamma l})} \right]$$

$$Z(l) = Z_0 \left[\frac{Z_L \cosh(\gamma l) + Z_0 \sinh(\gamma l)}{Z_0 \cosh(\gamma l) + Z_L \sinh(\gamma l)} \right]$$

$$Z(l) = Z_0 \left[\frac{Z_L + Z_0 \tanh(\gamma l)}{Z_0 + Z_L \tanh(\gamma l)} \right] \text{ Imp.}$$

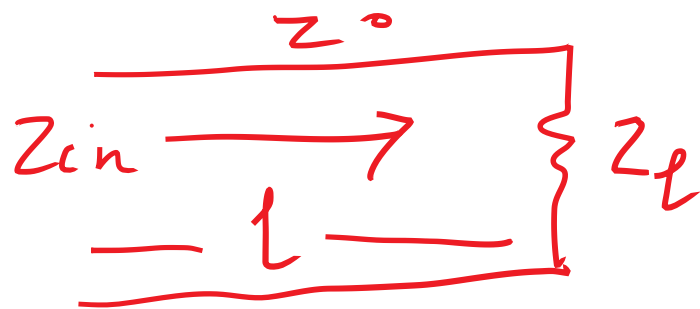
Input impedance at any point along a general (lossy) transmission line

Wave characteristic of finite Transmission line



$$Z_{in} = Z_o \left[\frac{Z_L + jZ_o \tan \beta l}{Z_o + jZ_L \tan \beta l} \right]$$

for lossless transmission line $\alpha = 0$



$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o}$$

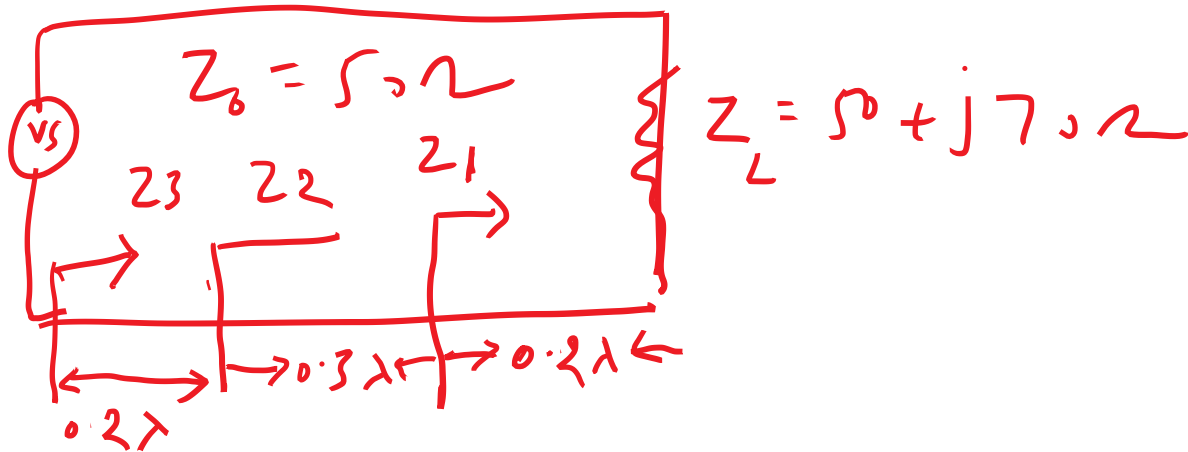
$$\Gamma(l) = \Gamma_L e^{-2\gamma l}$$

$$Z(l) = Z_o \left[\frac{1 + \Gamma(l)}{1 - \Gamma(l)} \right]$$

$$Z_{in} = Z_o \left[\frac{Z_L + Z_o \tanh \gamma l}{Z_o + Z_L \tanh \gamma l} \right]$$

βl ??

Example-1



Find out $Z_1 = ?$

$Z_2 = ?$

$Z_3 = ?$

Reflection Coefficient & VSWR



$$\Gamma(l) = \frac{Z(l) - Z_0}{Z(l) + Z_0} \quad \text{at } l=0$$
$$\Gamma_l = \frac{Z_l - Z_0}{Z_l + Z_0}$$
$$\Gamma_l = \frac{V_0^-}{V_0^+} \quad \Gamma(l) = \frac{V_0^-}{V_0^+} e^{-2\gamma l}$$

Measure of the mismatch of a line can be done by calculating standing wave ratio (SWR)-s

$$S = \frac{|V_{\max}|}{|V_{\min}|} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad (\text{Dimensionless}).$$

$$|\Gamma| = \frac{S - 1}{S + 1} \quad (\text{Dimensionless}).$$

Reflection Coefficient & VSWR



$$\Gamma = \frac{Z_L - Z_o}{Z_L + Z_o}$$

$$|\Gamma| = \frac{S - 1}{S + 1}$$

$$\left[S = \frac{1 + |\Gamma|}{1 - |\Gamma|} \right]$$

$\Gamma = 0, \quad S = 1$ when $Z_L = Z_o$ (Matched load);

$\Gamma = -1, \quad S \rightarrow \infty$ when $Z_L = 0$ (Short circuit);

$\Gamma = +1, \quad S \rightarrow \infty$ when $Z_L \rightarrow \infty$ (Open circuit).

what is value of $|\Gamma|$??
if load is inductive or capacitive

$$V = V^+ e^{j\beta l} + V^- e^{-j\beta l} \text{ lossless line}$$

$$V = V^+ e^{j\beta l} + V^- e^{-j\beta l}$$

$$V = V^+ e^{j\beta l} \left(1 + \frac{V^-}{V^+} e^{-j2\beta l} \right)$$

$$V = V^+ e^{j\beta l} \left(1 + \Gamma_L e^{-j2\beta l} \right)$$

$$V = V^+ e^{j\beta l} \left(1 + |\Gamma_L| e^{j\phi} e^{-j2\beta l} \right)$$

$$|V| = |V^+| \left| 1 + |\Gamma_L| e^{j(\phi - 2\beta l)} \right|$$

$$\text{for } V_{\max} \rightarrow |e^{j(\phi - 2\beta l)}| = 1$$

$I_{\min}$

$$\text{For } V_{\min}: \\ |e^{j(\phi - 2\beta l)}| = -1$$

$$\phi - 2\beta l = (2n+1)\pi$$

$I_{\max}$

$$\text{So } \phi - 2\beta l = 2n\pi$$

Rmax and Rmin



$$V_{max} = V^+ (1 + \Gamma_L)$$

$$I_{min} = \frac{V^+}{Z_0} (1 - \Gamma_L)$$

$$V_{min} = V^+ (1 - \Gamma_L)$$

$$I_{max} = \frac{V^+}{Z_0} (1 + \Gamma_L)$$

$$\text{So } R_{max} = Z_0 \left(\frac{1 + \Gamma_L}{1 - \Gamma_L} \right)$$

$$R_{min} = \frac{Z_0}{\left(\frac{1 + \Gamma_L}{1 - \Gamma_L} \right)}$$

$$\text{So } \underline{R_{max} = Z_0 (VSWR)}$$

$$R_{min} = \frac{Z_0}{VSWR}$$

Example



$$Z_{in} = \frac{Z_0 (Z_L + j Z_0 \tan \beta L)}{(Z_0 + j Z_L \tan \beta L)}$$

a) find Z_{in} at $l = \frac{\lambda}{2}$ and $l = \frac{\lambda}{4}$?

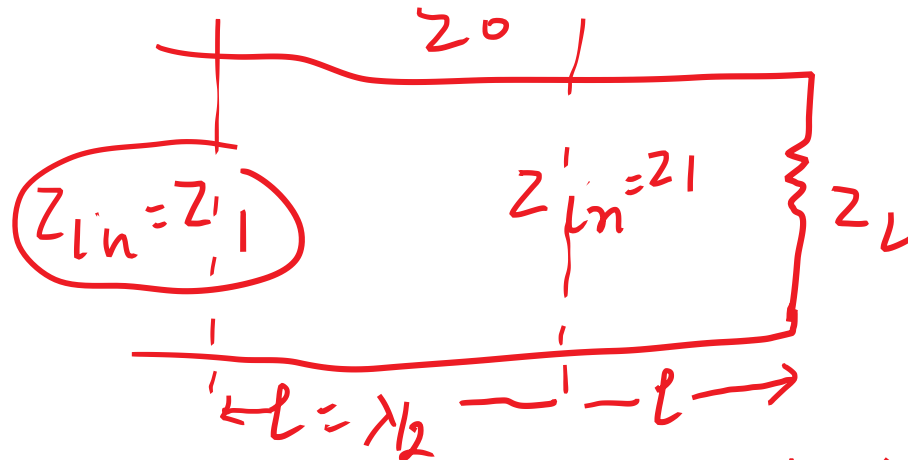
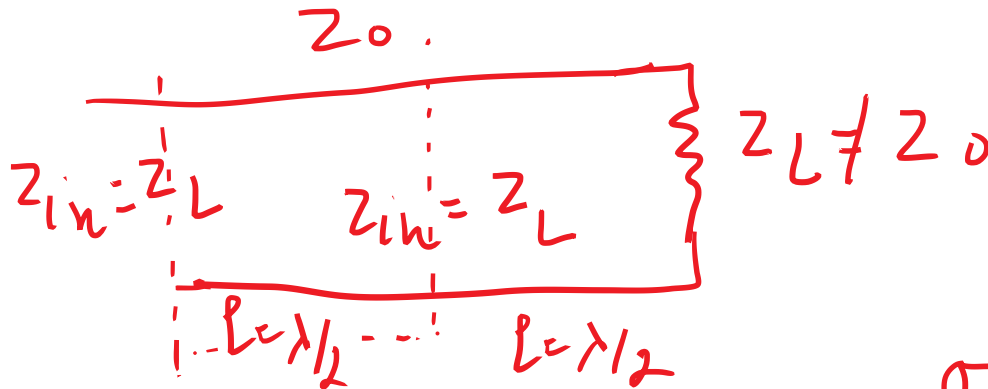
at $l = \lambda/2$

$$Z_{in} = Z_L$$

at $l = \lambda/4$

$$Z_{in} = \frac{Z_0^2}{Z_L}$$

Example



at $l = \lambda/2$

$$Z_{in} = Z_L$$

$\Rightarrow$ Impedance repeats at every distance $= \frac{\lambda}{2}$

Example-2



$$Z_0 = 50 \Omega$$

$$Z_L = 75 - 75j \Omega$$

Calculate a) Γ_L , VSWR

$$\text{VSWR} = 3.34$$

$$\Gamma_L = 0.54 \angle -40.6^\circ$$

$$\Gamma(0.2\lambda) = 0.54 \angle +175.4^\circ$$

$$Z(0.2\lambda) = (14.95 + j1.82) \Omega$$

$$\Gamma(0.3\lambda) = 0.54 \angle 103.39^\circ$$

$$Z(0.3\lambda) = (22.83 + j34.136) \Omega$$

b) $Z_{in}(0.2\lambda)$, $\Gamma_{in}(0.2\lambda)$

c) $Z_{in}(0.3\lambda)$, $\Gamma_{in}(0.3\lambda)$

d) $Z_{in}(0.7\lambda)$, $\Gamma_{in}(0.7\lambda)$

e) Location of V_{max} & V_{min} -

Example-2



Location of $V_{\max} \approx I_{\min} \approx R_{\max}$

$$\phi - 2\beta l = 2n\pi \quad n=1$$

$$-40.6^\circ - 2\beta l = 360^\circ$$

$$-2\beta l = 400.6^\circ$$

$$-l = \frac{400.6}{2 \times 2 \times 180} \lambda$$

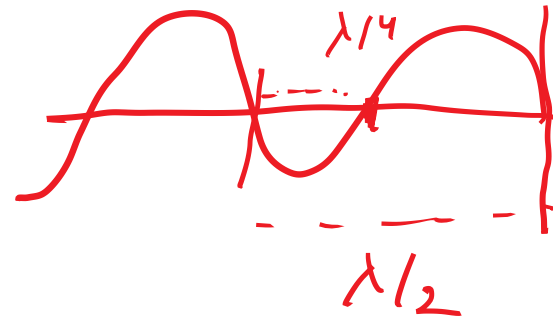
$$l = \underline{-0.55\lambda} + \underline{0.5\lambda} + \underline{0.5\lambda} = \underline{0.45\lambda}$$

maxima ↑

minima

impedance repeats at every $\lambda/2$ length

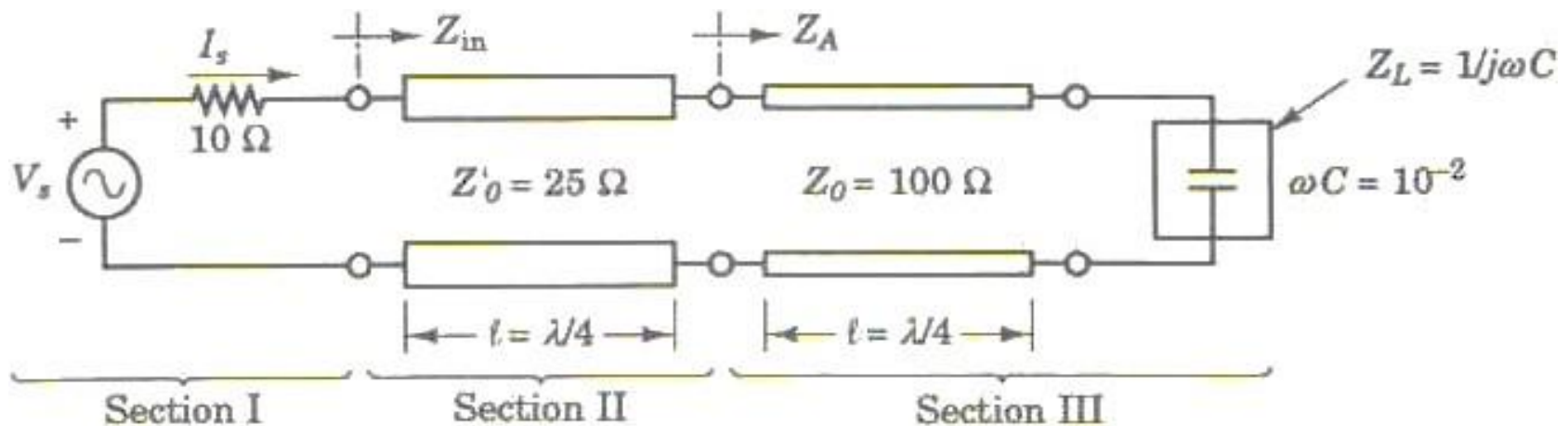
$$\rightarrow 0.45\lambda - 0.25\lambda = \boxed{0.20\lambda}$$

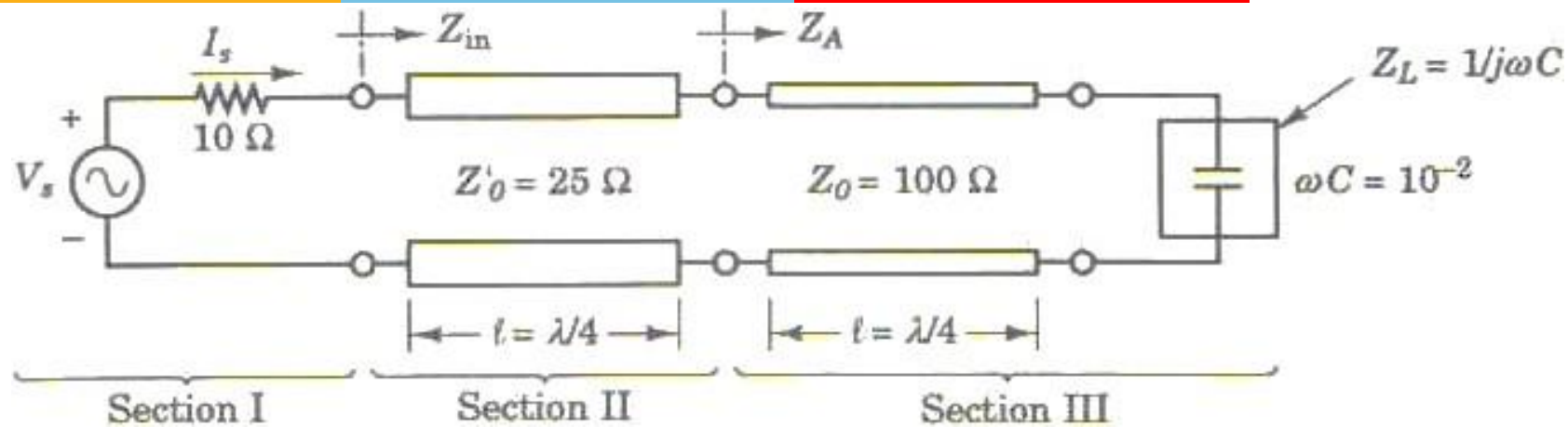


Example-2



What is the input impedance supplied by the source?





$$Z_i = Z_o \left[\frac{Z_L + jZ_o \tan \beta l}{Z_o + jZ_L \tan \beta l} \right]$$

$$Z_{\lambda/4} = \frac{Z_o^2}{Z_L}$$

Transmission Lines as circuit elements



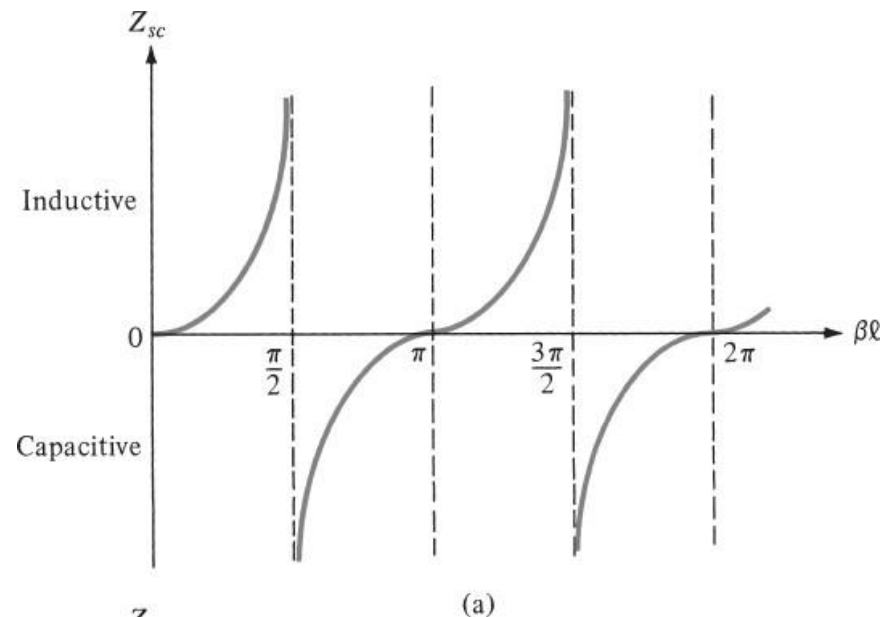
For a lossless line:

$$Z_{in} = Z_o \left[\frac{Z_L + jZ_o \tan \beta l}{Z_o + jZ_L \tan \beta l} \right]$$

Transmission Lines terminated in short or open circuits are commonly used as tuning elements.

If the line is **short-circuited**: $Z_L = 0$

$$Z_{SC} = jZ_o \tan \beta l$$



Transmission Lines as circuit elements



$$Z_{sc} = jZ_0 \tan \beta l$$

$$\text{if } \beta l < \pi/2 \\ l < \frac{\lambda}{4}$$

$$\tan \beta l \approx \beta l$$

$$Z_{sc} = jZ_0 \beta l$$

$$Z_{sc} = j\sqrt{\frac{L}{C}} \omega \sqrt{LC} \cdot l = j\omega L l = \underbrace{j\omega L}_{\text{Inductor}} \Rightarrow$$

$$\pi/2 < \beta l < \pi \rightarrow \text{Capacitor}$$

$$\text{at } \beta l = \pi/2 \rightarrow Z_{sc} = \infty \rightarrow \text{capacitor} \checkmark$$

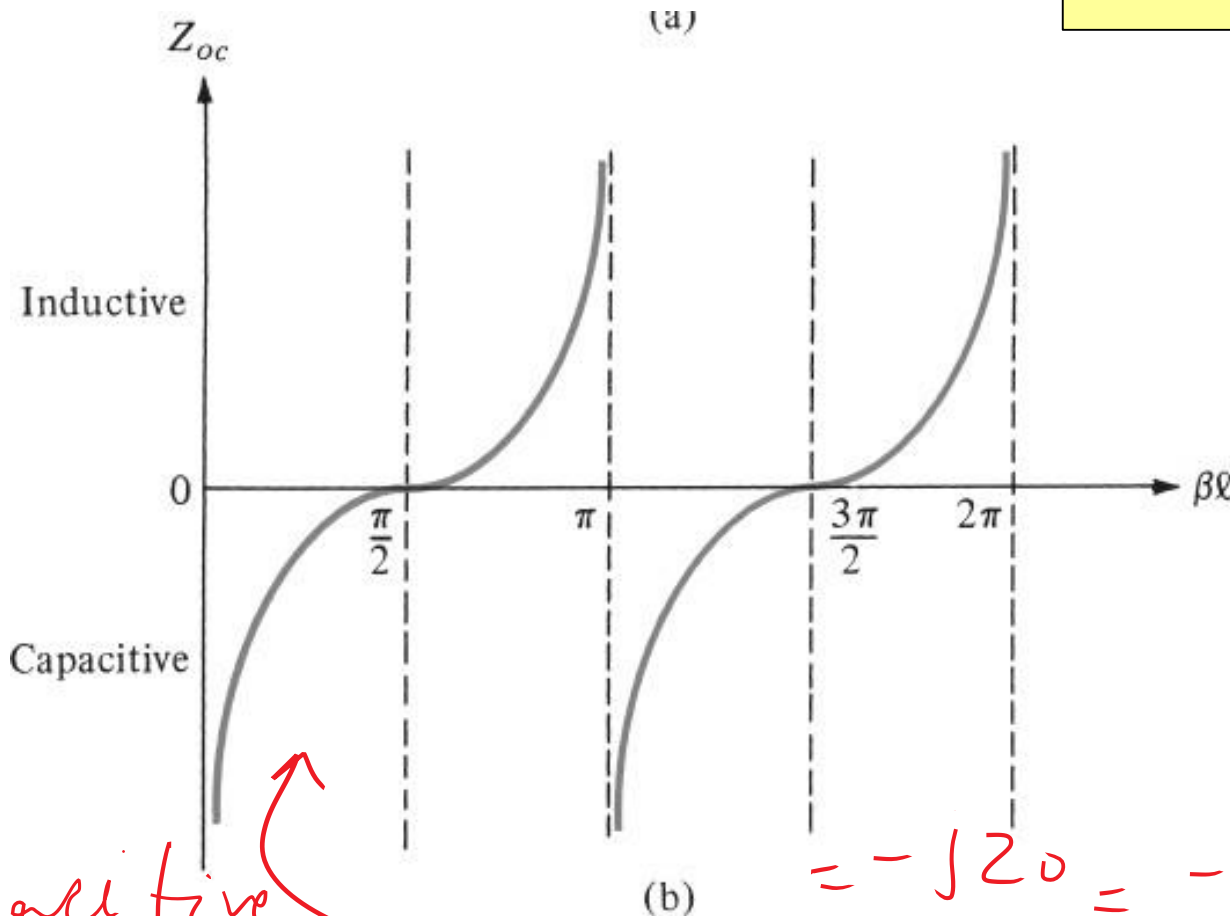
Transmission Lines as circuit elements



For a lossless line: open-circuited:

$$Z_i = Z_o \left\{ \frac{Z_L + jZ_o \tan \beta l}{Z_o + jZ_L \tan \beta l} \right\}$$

$$Z_{in} = -jZ_o \cot \beta l$$



$$\beta l < \frac{\pi}{2}$$

$$l < \frac{\lambda}{4}$$

$$= -jZ_o \tan \beta l$$

$$= -jZ_o = -j\sqrt{L/C} = \left[\frac{1}{j\omega C} \right]$$

capacitive

Transmission Lines as circuit elements



$$Z_{sc} = jZ_0 \tan \beta l$$

$$Z_{oc} = -jZ_0 \cot \beta l = \frac{Z_0}{j \tan \beta l}$$

$$Z_0 = \sqrt{Z_{oc} Z_{sc}}$$

$$Z_0 = \sqrt{Z_{OC} Z_{SC}}$$
$$\gamma = \frac{1}{l} \tanh^{-1} \sqrt{\frac{Z_{SC}}{Z_{OC}}}$$

$$\frac{Z_{sc}}{Z_{oc}} = \tan^2 \beta l$$

$$\beta = \frac{1}{l} \tan^{-1} \sqrt{\frac{Z_{sc}}{Z_{oc}}}$$

Lossy line